Numerical Analysis Assignment 1

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Assessed exercise 1.1 (Lipschitz condition)

1. Suppose that $g\colon D \to \mathbb{R}$ is differentiable with g' bounded in D. Show that g is Lipschitz differentiable. *Hint:* Use the mean value theorem in integral form:

$$g(y) - g(x) = (y - x) \int_0^1 g'(x + s(y - x))\, ds, \tag{1}$$

and basic properties of the integral:

$$\left| \int f \right| \leq \int |f|, \tag{2}$$

$$f \leq h \implies \int f \leq \int h. \tag{3}$$

SOLUTION:

By the MVT in integral form:

$$g(y)-g(x) = (y-x) \int_0^1 g'(x+s(y-x))\, ds \iff \frac{g(y) - g(x)}{y - x} = \int_0^1 g'(x+s(y-x))\, ds$$

Taking norms:

$$\frac{|g(y) - g(x)|}{|y - x|} = \left| \int_0^1 g'(x + s(y - x))\, ds \right|$$

As hinted:

$$\left| \int_0^1 g'(x + s(y - x))\, ds \right| \leq \int_0^1 |g'(x + s(y - x))|\, ds$$

Now because know g is differentiable and g' is bounded in D, that means $\forall x \in D$, $g'(x)$ exists and $\exists L > 0$ such that $|g'(x)| \leq L := \sup_{x \in D} |g'(x)|$. Assuming D is convex so $x + s(y - x) \in D\, \forall s \in [0, 1]$, the above yields:

$$\int_0^1 |g'(x + s(y - x))|\, ds \leq \int_0^1 L\, ds = L$$

So we have shown that there exists some $L > 0$ such that $\forall x, y \in D$:

$$\frac{|g(y) - g(x)|}{|y - x|} \leq L \iff |g(y) - g(x)| \leq L\, |y - x|$$

Which proves that g is Lipschitz on D. $\square$

2. Deduce that if $g \in C^1(D)$ and C is a compact, (i.e. closed and bounded) **convex**, subset of D, **with non-empty interior** then $g|_C \in \mathrm{Lip}(C)$ and

$$|g|_{\mathrm{Lip}\, C} = \sup_{x \in C} \|g'(x)\| = \max_{x \in C} \|g'(x)\|. \tag{4}$$

(The last equality is a formulaic way of saying that the least upper bound is achieved by some point, i.e. $\exists\, x^* \in C$ with $\|g'(x^*)\| = \sup_{x \in C} \|g'(x)\|$.)

SOLUTION:

We are given that $g : D \to Y \; : \; g \in C^1(D)$ so g' is continuous on D and thus $\|g'\| \in C^0(D) \implies \|g'\| \in C^0(C)$.

Since $C \subset D$ is compact and $\|g'\| \in C^0(C)$, by the Extreme Value Theorem, we have that $\|g'\|$ is bounded on C and therefore,

$$\exists x^* \in C \text{ with } \|g'(x^*)\| = \sup_{x \in C} \|g'(x)\| = \max_{x \in C} \|g'(x)\| := M < \infty.$$

I.e. $\|g'\|$ attains its maximum on C. From this we can apply our previous result:

$$\frac{\|g(y) - g(x)\|}{\|y - x\|} \leq \int_0^1 \|g'(x + s(y - x))\| \, ds$$

Since we have assumed that C is convex, we have $x + s(y - x) \in C \, \forall \, s \in [0, 1]$

$$\implies \int_0^1 \|g'(x + s(y - x))\| \, ds \leq \int_0^1 M \, ds = M$$

$$\implies \frac{\|g(y) - g(x)\|}{\|y - x\|} \leq M \; \forall x \neq y \in C$$

Therefore, $g|_C \in \text{Lip}(C)$ and hence:

$$\sup_{x \neq y \in C} \frac{\|g(x) - g(y)\|}{\|x - y\|} = \left|g_{|C}\right|_{\text{Lip}(C)} := L \leq M = \max_{x \in C} \|g'(x)\| = \sup_{x \in C} \|g'(x)\|$$

Now because we have shown g to be Lipschitz, it holds that

$$\frac{\|g(y) - g(x)\|}{\|y - x\|} \leq L$$

Next we can fix some $x, y \in C^{\text{o}}$, $x \neq y$ and define $y_t := x + t(y - x)$ with $t \in (0, 1]$ to ensure $x \neq y_t \, \forall \, t$. I.e. we let $y_t \in C^{\text{o}}$ to be the line segment between any two distinct values x and y in the interior of C. Note that by convexity of C, $y_t \in C \, \forall t \in (0, 1]$. Because we have shown g to be Lipschitz, it holds that:

$$\frac{\|g(y_t) - g(x)\|}{\|y_t - x\|} = \frac{\|g(x + t(y - x)) - g(x)\|}{\|x + t(y - x) - x\|} = \frac{\|g(x + t(y - x)) - g(x)\|}{t\|y - x\|} \leq L$$

Since we have that $g \in C^1(D)$, we can take $\lim_{t \to 0^+}$ giving:

$$\left\| g'(x) \left[\frac{y - x}{\|y - x\|} \right] \right\| \leq L$$

Now, for every $x, y \in C^{\text{o}}$, taking the supremum over all directions of the unit vector $h := \frac{y - x}{\|y - x\|}$ gives:

$$\sup_{\|h\| = 1} \|g'(x)[h]\| = \|g'(x)\| \leq L \quad \forall x \in C^{\text{o}}$$

$$\implies \sup_{x \in C^{\text{o}}} \|g'(x)\| \leq L$$

Then, fixing $x \in \partial C$ and choosing any $\tilde{x} \in C^{\text{o}}$, we define $x_k := \tilde{x} + k(x - \tilde{x}); k \in [0, 1)$ with $x_k \in C^{\text{o}} \, \forall k \in [0, 1)$ by convexity of C. Then because $x_k \to x$ as $k \to 1^-$, from the above interior case and continuity of $\|g'\|$ on C, we have that:

$$\|g'(x_k)\| \leq L \quad \forall x \in \partial C, k \in [0, 1)$$

$$\implies \|g'(x)\| = \lim_{k \to 1^-} \|g'(x_k)\| \leq L \quad \forall x \in \partial C$$

$$\implies \sup_{x\in\partial C} \|g'(x)\| \leq L$$

Combining the above results yields $\sup_{x\in C} \|g'(x)\| \leq L$ since $\partial C \cup C^{\circ} = C$.

Again combining this with our previously obtained inequality gives:

$$\sup_{x\in C} \|g'(x)\| \leq |g|_{\mathrm{Lip}(C)} = \sup_{x\in C} \|g'(x)\| = \max_{x\in C} \|g'(x)\|$$
$$\implies |g|_{\mathrm{Lip}(C)} = \sup_{x\in C} \|g'(x)\| = \max_{x\in C} \|g'(x)\| \quad \square$$

3. Find a function f which is Lipschitz continuous on $[-1, 1]$ but not differentiable at all interior points of $[-1, 1]$.

SOLUTION:

Our candidate is $f(x) = |x|$ which is notably not differentiable at $x = 0 \in [-1, 1]$. So we can now check whether $f(x) = |x| \in \mathrm{Lip}([-1, 1])$.

$$|f(x) - f(y)| = ||x| - |y|| \leq |x - y| \quad \text{by the triangle inequality.}$$
$$\implies |f(x) - f(y)| \leq 1 \cdot |x - y| \,\forall\, x, y \in [-1, 1]$$
$$\implies f(x) = |x| \in \mathrm{Lip}([-1, 1]), f \notin C^1([-1, 1]).$$

So, our candidate function, $f(x) = |x|$ meets the given criteria. $\square$

Assessed exercise 1.2 (Solving a matrix equation with fixed point theory)

Let $A \in \mathbb{R}^{n\times n}$ be an invertible matrix. Suppose that $a_{ii} \neq 0$ for $i = 1, \ldots, n$. Let $D, R \in \mathbb{R}^{n\times n}$ be a decomposition of $A = D + R$ into diagonal part D

$$D = \begin{pmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & a_{nn} \end{pmatrix}. \tag{5}$$

and remainder R

$$R = \begin{pmatrix} 0 & a_{12} & \cdots & a_{1n} \\ a_{21} & 0 & \ddots & \vdots \\ \vdots & \ddots & \ddots & a_{n-1,n} \\ a_{n1} & \cdots & a_{n,n-1} & 0 \end{pmatrix}. \tag{6}$$

Let $b \in \mathbb{R}^n$ be a given vector. Given an initial $x^{(0)} \in \mathbb{R}^n$ the Jacobi iteration constructs $x^{(k+1)}$ by recursively defining

$$x^{(k+1)} = D^{-1}\big(b - Rx^{(k)}\big). \tag{7}$$

1. In the fixed point iteration in (7), state the function f such that $x^{(k+1)} = f\big(x^{(k)}\big)$.

SOLUTION:

$$f(x) = D^{-1}\big(b - Rx\big)\,;\; f : \mathbb{R}^n \to \mathbb{R}^n \quad \square$$

Note that $a_{ii} \neq 0 \implies D^{-1}$ exists.

2. Supposing that the fixed point iteration converges, to what does it converge?

SOLUTION:

Under the supposition that the fixed point iteration converges, we have:

$$x^* = \lim_{k\to\infty} x^{(k+1)} = \lim_{k\to\infty} f(x^{(k)}) = f(x^*) = D^{-1}\big(b - Rx^*\big)$$

$$x^* = D^{-1}\big(b - Rx^*\big) \implies Dx^* = b - Rx^*$$
$$\implies Dx^* + Rx^* = b \implies (D+R)x^* = b$$
$$\iff Ax^* = b \implies x^* = A^{-1}b\,;\, x^* \in \mathbb{R}^n \quad \square$$

Note, we are given that A is invertible so x^* is well-defined and unique.

We can notice that the Jacobi iteration provides an iterative method to solve $Ax = b$ where it is potentially "expensive" to invert A.

3. State a sufficient condition on the matrices R and D such that the iteration will converge. Hint: you might want to use Banach–Caccioppoli's theorem.

SOLUTION:

We know under Banach–Caccioppoli's theorem that $f(x) = D^{-1}\big(b - Rx\big)$ will have a **unique** fixed point x^* if:

(a) $f : X \to \mathbb{R}^n$ is a contraction on some closed subset $X \subset \mathbb{R}^n$

(b) $x^* \in X$ such that $f(x^*) = x^*$

The condition for f to be a contraction on X is $\exists \kappa \in [0,1)$ such that:

$$\|f(x) - f(y)\| \leq \kappa \|x - y\| \;\forall\, x, y \in X.$$

We have:

$$\|f(x) - f(y)\| = \|D^{-1}(b - Rx) - D^{-1}(b - Ry)\|$$
$$= \|D^{-1}R(y - x)\| = \|D^{-1}R(x - y)\| \leq \|D^{-1}R\| \cdot \|x - y\|$$

Which gives us that:

$$\|f(x) - f(y)\| \leq \|D^{-1}R\| \cdot \|x - y\|$$

Therefore, f is a contraction on $X \iff \kappa := \|D^{-1}R\| < 1$ (Note that $a_{ii} \neq 0 \implies D^{-1}$ exists)

Since this inequality holds $\forall\, x, y \in \mathbb{R}^n$ which is closed in of itself, taking $X = \mathbb{R}^n$, we have a sufficient condition for f to be a contraction on $\mathbb{R}^n$.

We have already shown in 1.2.2 that $\exists\, x^* = A^{-1}b \in \mathbb{R}^n$ such that $f(x^*) = x^*$. So by Banach-Cacciopolli, x^* is the unique fixed point of f.

Therefore, the only condition necessary to ensure this iteration will converge further to the established condition $\|D^{-1}R\| < 1$ is ensuring $x_0 \in X = \mathbb{R}^n$ which is trivially satisfied since $dom(f) = \mathbb{R}^n$. $\square$

Assessed exercise 1.3 (Super linear Banach-Caccioppoli)

1. Suppose $g : D \to D$ is a contraction in a closed set $D \subset \mathbb{R}$ with fixed point x^*. Suppose also that $g \in C^1(D)$ and g' satisfies that there is $C > 0$ and $\alpha \in (0,1)$ such that

$$\|g'(x) - g'(y)\| \leq C\|x - y\|^\alpha \quad \text{for all } x, y \in D.$$

If $g'(x^*) = 0$, then the convergence of the iteration $x_{k+1} = g(x_k)$ is super-linear with order $1 + \alpha$. Prove this statement.

SOLUTION:

We begin by denoting $e_k := x_k - x^*$ and $\epsilon_k = |e_k|$.

We can also take note that because x^* is a fixed point of g, we know $g(x^*) = x^*$.

Now, by the mean value theorem in integral form, we have:

$$\epsilon_{k+1} = |x_{k+1} - x^*| = |g(x_k) - g(x^*)| = \left|(x_k - x^*)\int_0^1 g'(x^* + s(x_k - x^*))ds\right|$$

$$\leq |x_k - x^*| \int_0^1 |g'(x^* + s(x_k - x^*))|\, ds$$

Because we know $\exists C > 0$, $\alpha \in (0,1)$, s.t.:

$$\|g'(x) - g'(y)\| \leq C\|x - y\|^{\alpha} \quad \forall x, y \in D. \text{(i.e. } g' \text{ is } \alpha\text{-Holder)}$$

Then if $g'(x^*) = 0$:

$$\implies |g'(x^* + s(x_k - x^*))| = |g'(x^* + s(x_k - x^*)) - g'(x^*)|$$

$$\leq C|\,(x^* + s(x_k - x^*)) - (x^*)\,|^{\alpha} = Cs^{\alpha}|\,((x_k - x^*))\,|^{\alpha}$$

Returning to our integral expression:

$$\epsilon_{k+1} \leq |x_k - x^*| \int_0^1 |g'(x^* + s(x_k - x^*))|\, ds = |x_k - x^*| \int_0^1 |g'(x^* + s(x_k - x^*)) - g'(x^*)|\, ds$$

$$\leq \|x_k - x^*\| \int_0^1 |(Cs^{\alpha}|\,(x_k - x^*)\,|^{\alpha})|\, ds = \frac{C}{1+\alpha}|x_k - x^*|^{1+\alpha} = \frac{C}{1+\alpha}\epsilon_k^{1+\alpha}$$

At this point we have shown that:

$$\epsilon_{k+1} \leq \frac{C}{1+\alpha}\epsilon_k^{1+\alpha}$$

However, to show that the convergence of this iteration is superlinear with rate $\alpha+1$, we need to that $\{x_k\}_{k\in\mathbb{N}}$ does indeed converge (to x^*). We are given that:

(a) $D \subset \mathbb{R}$ is closed

(b) $g : D \to D$ is a contraction,
i.e. $\exists \kappa \in [0,1)$ with $|g(x) - g(y)| \leq \kappa\,|x - y| \quad \forall x, y \in D \iff |g|_{\mathrm{Lip}(D)} = \kappa$

(c) g has a fixed point, x^*.

Therefore, by Banach's Fixed Point Theorem, g has exactly one fixed point, which is unique, namely $x^* \in D$.

Then, provided we have chosen $x_0 \in D$, given we defined $x_{k+1} = g(x_k)$, we know by BFPT that $x_k \to x^*$ as $k \to \infty$.

Hence, the convergence of the iteration $x_{k+1} = g(x_k)$ is super-linear with order $1 + \alpha$ and consequently, the statement holds. $\square$
